

coding

gate
TC, SC

CS, IT

gate CSE
gate DA

Lec 46

Given an Array of n distinct elements (Integers), and an integer -k
, what will be worst case time complexity to find kth smallest
element?

int a[n];

int k

array an min element = 1st smallest element -

Ex: ip /op IP: $[10, 50, 6, 14, 3, 2]$, $k=3$

OP: 3rd smallest element = 6

Ans = 6

Brute force
 $k=3$

IF array is sorted

$O(1)$
return $a[k-1]$

$[2, 3, 6, 10, 14, 50]$
 $0 \ 1 \ 2 \ 3 \ 4 \ 5$

return $k-i$ th index that element

Brute force

If random array is given

Sort it

$\Rightarrow n \log n$

Best algorithm and worst-case TC

By merge sort

$\Rightarrow n \log n$

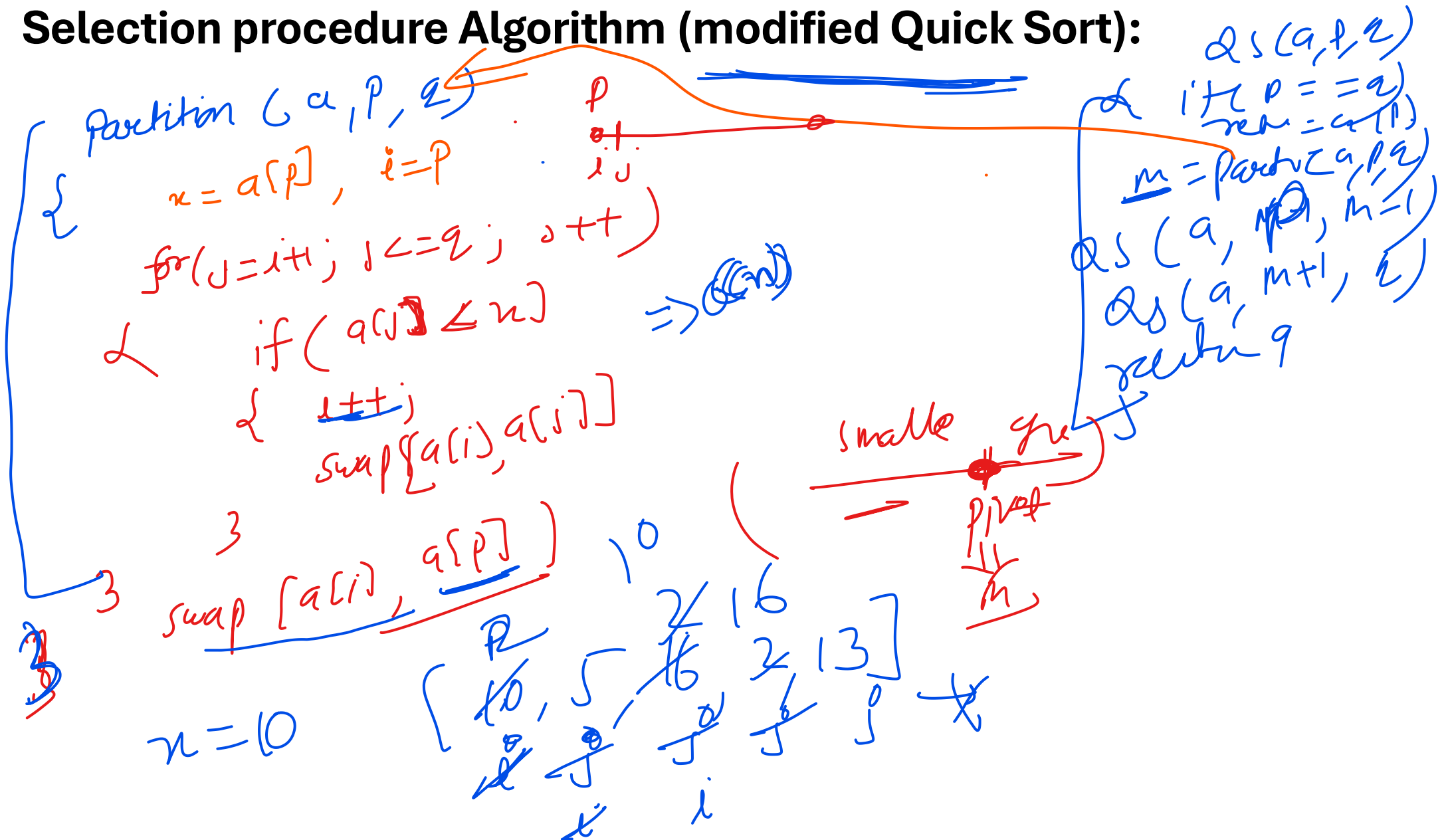
~~$n \log n$~~
 ~~$n \log n$~~

return arr

$\Rightarrow n \log n + O(1)$

Modified merge also $\Rightarrow$ Intervall

Selection procedure Algorithm (modified Quick Sort):



$[2, 5, 10, 16, 3, 4, 13]$
 $0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6$
 pivot is at correct place

$if \ m = l - 1$

arr

$l = 4$
 arr

$l = 2$

arr

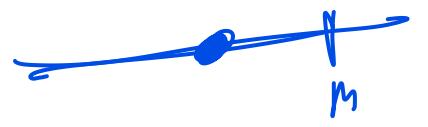
of
selectn
problem

Algo:

```

sp(a, p, q, k)
{
  if (p == q) return a(p)
  m = partition(a, p, q)
  if (m == k-1)
    return a(m) or a[k-1]
  if (m > k-1)
    return partition(a, p, m-1)
  else
    return partition(a, m+1, q)
}
  
```

$O(n)$

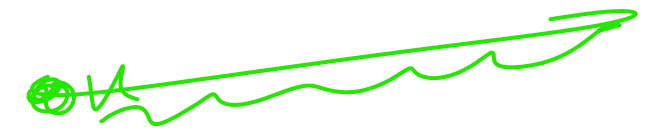


In Best case: $T(n) = T(\frac{n}{2}) + n$

$\Rightarrow O(n)$

$n \mid n^{\log_6 9}$
 $n \mid n^{\log_2 10}$

$n^{\log 6}$



for worst case:

$$T(n) = T(n-1) + n$$

$$O(n^2)$$

$$BC = O(n)$$
$$WC = O(n^2)$$

not in smallest
@ limit

SP (a, p, n, K)

$\int_m \rightarrow \text{pau}$

$\{ \Rightarrow \}$

$\{ \text{met} \}$

$\rightarrow \text{pau}$

